

Full Associative Cache Mapping

Sequential Sequence: [if cache block = 4] 0,1,2,3,4,5,6,7,0,1,2,3,4,5,6,7

Block Size: 4 words/block

Cache Access Time, T_c : 1ns

Memory Access Time, T_m : 100ns

Least Recently Used (LRU):

# of Accesses	Data	Block #0	Block #1	Block #2	Block #3
1	0	0	-	-	-
2	1	0	1	-	-
3	2	0	1	2	-
4	3	0	1	2	3
5	4	4	1	2	3
6	5	4	5	2	3
7	6	4	5	6	3
8	7	4	5	6	7
9	0	0	5	6	7
10	1	0	1	6	7
11	2	0	1	2	7
12	3	0	1	2	3
13	4	4	1	2	3
14	5	4	5	2	3
15	6	4	5	6	3
<u>16</u>	<u>7</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>

AMAT = $hc + (1-hc)*MP$, where

$hc = T_c * \text{hit_rate}$

$MP = T_c + \text{block_size} * T_m + T_c$

AMAT = 402 ns

$T_{AT} = (\#_{\text{hits}} * \text{block_size}) * T_c + (T_c + T_m * \text{block_size} + T_c * \text{block_size}) * \#_{\text{misses}}$

$T_{AT} = 6480 \text{ ns}$

of hits: 0

of misses: 16

hit rate: 0.00

miss rate: 100.0

Block Size: 4 words/block

Cache Access Time, T_c : 1ns

Memory Access Time, T_m : 100ns

Total Access Time, T_{AT}

Sequential Sequence: [if cache block = 4] 0,1,2,3,4,5,6,7,0,1,2,3,4,5,6,7

Block Size: 4 words/block

Cache Access Time, T_c : 1ns

Memory Access Time, T_m : 100ns

Most Recently Used (MRU):

# of Accesses	Data	Block #0	Block #1	Block #2	Block #3
1	0	0	-	-	-
2	1	0	1	-	-
3	2	0	1	2	-
4	3	0	1	2	3
5	4	0	1	2	4
6	5	0	1	2	5
7	6	0	1	2	6
8	7	0	1	2	7
9	0	0	1	2	7
10	1	0	1	2	7
11	2	0	1	2	7
12	3	0	1	3	7
13	4	0	1	4	7
14	5	0	1	5	7
15	6	0	1	6	7
16	7	0	1	6	7

AMAT = $hc + (1-hc)*MP$, where

$hc = T_c * \text{hit_rate}$

$MP = T_c + \text{block_size} * T_m + T_c$

AMAT = 301.75 ns

$T_{AT} = (\#_{\text{hits}} * \text{block_size}) * T_c + (T_c + T_m * \text{block_size} + T_c * \text{block_size}) * \#_{\text{misses}}$

$T_{AT} = 4876 \text{ ns}$

of hits: 4

of misses: 12

hit rate: 4/16

miss rate: 12/16

Block Size: 4 words/block

Cache Access Time, T_c : 1ns

Memory Access Time, T_m : 100ns

Total Access Time, T_{AT}

Mid-repeat Blocks: [if cache block = 4] 0,1,2,3,1,2,3,4,5,6,7, 0,1,2,3,1,2,3,4,5,6,7

Block Size: 4 words/block

Cache Access Time, T_c : 1ns

Memory Access Time, T_m : 100ns

Least Recently Used (LRU):

# of Accesses	Data	Block #0	Block #1	Block #2	Block #3
1	0	0	-	-	-
2	1	0	1	-	-
3	2	0	1	2	-
4	3	0	1	2	3
5	1	0	1	2	3
6	2	0	1	2	3
7	3	0	1	2	3
8	4	4	1	2	3
9	5	4	5	2	3
10	6	4	5	6	3
11	7	4	5	6	7
12	0	0	5	6	7
13	1	0	1	6	7
14	2	0	1	2	7
15	3	0	1	2	3
16	1	0	1	2	3
17	2	0	1	2	3
18	3	0	1	2	3
19	4	4	1	2	3
20	5	4	5	2	3
21	6	4	5	6	3
22	7	4	5	6	7

AMAT = hc + (1-hc)*MP, where

hc = T_c * hit_rate

MP = T_c + block_size * T_m + T_c

AMAT = 292.64 ns

$T_{AT} = (\#_{hits} * block_size) * T_C + (T_c + T_m * block_size + T_c * block_size) * \#_{misses}$

$T_{AT} = 6508$ ns

of hits: 6

of misses: 16

hit rate: 6/22

miss rate: 16/22

Block Size: 4 words/block

Cache Access Time, T_c : 1ns

Memory Access Time, T_m : 100ns

Total Access Time, T_{AT}

Mid-repeat Blocks: [if cache block = 4] 0,1,2,3,1,2,3,4,5,6,7, 0,1,2,3,1,2,3,4,5,6,7

Block Size: 4 words/block

Cache Access Time, T_c : 1ns

Memory Access Time, T_m : 100ns

Most Recently Used (MRU):

# of Accesses	Data	Block #0	Block #1	Block #2	Block #3
1	0	0	-	-	-
2	1	0	1	-	-
3	2	0	1	2	-
4	3	0	1	2	3
5	1	0	1	2	3
6	2	0	1	2	3
7	3	0	1	2	3
8	4	0	1	2	4
9	5	0	1	2	5
10	6	0	1	2	6
11	7	0	1	2	7
12	0	0	1	2	7
13	1	0	1	2	7
14	2	0	1	2	7
15	3	0	1	3	7
16	1	0	1	3	7
17	2	0	2	3	7
18	3	0	1	3	7
19	4	0	1	4	7
20	5	0	1	5	7
21	6	0	1	6	7
22	7	0	1	6	7

AMAT = hc + (1-hc)*MP, where

hc = T_c * hit_rate

MP = T_c + block_size * T_m + T_c

AMAT = 237.95 ns

$T_{AT} = (\#_{hits} * block_size) * T_c + (T_c + T_m * block_size + T_c * block_size) * \#_{misses}$

$T_{AT} = 5301$ ns

of hits: 9

of misses: 13

hit rate: 9/22

miss rate: 13/22

Block Size: 4 words/block

Cache Access Time, T_c : 1ns

Memory Access Time, T_m : 100ns

Total Access Time, T_{AT}